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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	Qart.ai
Project / Solution Name	Qart.ai
Name (Leader)	Kydyrgaliyev Meiirman
Email & Phone (Leader)	meirman02.07@gmail.com · +60 11-2725 6756
University Affiliation(s)	Xiamen University Malaysia
Live Prototype URL	https://qartai.replit.app/start
Code Repository URL (public / access granted to judges)	https://github.com/Moo7x/qart.ai (public; Apache License 2.0)

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

The user works in a React web app served by a FastAPI backend on Replit Autoscale. A synthetic CV first passes a rule-based scrubber that removes names, IC numbers, phone numbers and emails. The scrubbed text goes to OpenAI gpt-4o-mini with a strict JSON schema; every skill claim must quote the CV, and code checks each quote. Job titles and free-text tasks go to a RunPod serverless GPU, where BGE-M3 embeds them and a BGE reranker orders the candidates; if the GPU does not answer within its time budget, BM25 keyword search takes over and is labelled.

All scores are computed in Python from stored evidence: fit, skills gap, scenario weights, the learning plan and the with/without-AI comparison. Profiles, plans, CV versions, feedback and dated evidence live in PostgreSQL (Replit production database) through one SQLAlchemy layer. A Replit Scheduled deployment runs the evidence pipeline daily against the same database. It collects seven public sources, validates dates and units, stores immutable observations and proposes updates that users accept or decline. gpt-4o-mini is also used, on request, to rephrase computed facts in plain words; sentences containing numbers or forecasts are rejected.

Diagram: Appendix A; proof of use: Appendix B.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.12	Backend, pipeline, tests	PSF License
TypeScript 6.0	Web app language	Apache-2.0
FastAPI 0.155.0 (with Starlette)	HTTP API	MIT (Starlette BSD-3)
Uvicorn 0.54.0	ASGI server on Replit	BSD-3-Clause

Name & Version	Purpose in Project	License
Pydantic 2.14.1	Request and data validation	MIT
SQLAlchemy 2.1.6 (Core)	One database layer for SQLite and PostgreSQL	MIT
psycopg 3.3.6 (binary)	PostgreSQL driver	LGPL-3.0 (used unmodified)
httpx 0.28.1	Calls to public data sources and RunPod	BSD-3-Clause
openai 3.36.1 (Python SDK)	Client for gpt-4o-mini	Apache-2.0
PyYAML 6.0.3	Data packs and source registry	MIT
pypdf 6.21.2	Reading synthetic PDF CVs	BSD-3-Clause
python-docx 1.2.0	Reading synthetic DOCX CVs	MIT
python-multipart 0.0.32	CV upload form parsing	Apache-2.0
React 19.3 + react-dom	User interface	MIT
react-router-dom 7.18	Page routing	MIT
Recharts 3.10	Trends and Overview charts	MIT
lucide-react 1.61	Icons	ISC
Vite 8.3, @vitejs/plugin-react	Web build	MIT
Tailwind CSS 4.3	Styling	MIT
sentence-transformers 3.4.1, transformers 4.51.3 (RunPod worker only)	Loading BGE-M3 and the reranker on the GPU	Apache-2.0

Name & Version	Purpose in Project	License
PyTorch 2.6 (pytorch/pytorch CUDA 12.4 image, RunPod worker only)	GPU runtime for the models	BSD-3-Clause
pytest 9.1.1, ruff 0.17.1, mypy 2.4.0, oxlint 1.88 (development)	Tests, lint, type checks	MIT
Geist and Geist Mono fonts (Google Fonts)	Typography	SIL Open Font License 1.1
SDC Hackathon submission template (organisers)	Repository starting point	Apache-2.0

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
OpenAI API — gpt-4o-mini	Read scrubbed CVs and job postings into quoted skill claims; plain-words role explanation	paid, the team's own key
RunPod Serverless API (our endpoint)	Embeddings and reranking for occupation and skill matching	sponsor-provided credits
OpenDOSM API (api.data.gov.my), DOSM Job Market Insights release pages	Malaysian employment and online job-ad evidence	free, public
Hugging Face dataset download (Anthropic Economic Index)	AI-use evidence per occupation and task	free, public
O*NET 31.0 database files (onetcenter.org)	Occupations, tasks and task importance	free, public
ILO WP140 score file (GitHub)	Generative-AI exposure per occupation	free, public
ESCO API v1 (European Commission)	ISCO-08 links for occupations	free, public
Microsoft Learn Catalog API	Verifying course titles, length and level (curation)	free, public
YouTube oEmbed endpoint	Verifying video title and channel (curation only, not at runtime)	free, public
Job boards (JobStreet, Indeed Malaysia, MYFutureJobs)	"See job posts" links; no API calls, no scraping	free; links only

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit Autoscale deployment	Hosts the web app and API (the live prototype)	sponsor-provided credits
Replit PostgreSQL (production database)	Profiles, plans, CV versions, evidence, proposals	included with the Repl
Replit Scheduled deployment	Runs the evidence pipeline daily into the same database	sponsor-provided credits · Asia · 1 vCPU / 2 GiB · daily at 01:00 MYT
RunPod Serverless endpoint (queue-based)	GPU inference for BGE-M3 and the reranker; worker image built from the GitHub repository	sponsor-provided credits · GPU pools 24 GB and 16 GB (RTX 3090, L4, RTX A5000, RTX PRO 6000 MIG 24 GB) · max 1 worker, scales to zero
GitHub	Source code; RunPod builds the worker image from it	public repository
Proof of use	Screenshots in Appendix B; also submission/REPLIT PROOF OF USE.pdf and RUNPOD PROOF OF USE.pdf	—

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
gpt-4o-mini	OpenAI (API)	Pre-trained, called through the API with strict JSON schemas	OpenAI API terms
BAAI/bge-m3	Beijing Academy of AI, via Hugging Face; hosted on our RunPod endpoint	Pre-trained, inference only (no fine-tuning)	MIT
BAAI/bge-reranker-v2-m3	Beijing Academy of AI, via Hugging Face; hosted on our RunPod endpoint	Pre-trained, inference only	Apache-2.0
BM25 keyword ranking, simple forecast baselines (no change, straight line, damped trend, smoothing)	Written by the team	Self-built; fallback search and the backtest that decides whether trends are shown	Our code (Apache-2.0)

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
OpenDOSM — employees by occupation (wages_occupation)	api.data.gov.my	public	Malaysian Government Open Data Terms 1.0	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
DOSM Big Data Analytics: Job Market Insights, Q2 2023 – Q2 2026	dosm.gov.my	public	Malaysian Government Open Data Terms 1.0	N
Anthropic Economic Index, release 2026-06-26	huggingface.co/datasets/Anthropic/EconomicIndex	public	CC BY	N
O*NET 31.0 Database	onetcenter.org	public	CC BY 4.0	N
ILO WP140 generative-AI exposure scores (Gmyrek et al., 2025)	github.com/pgmyrek/2025_GenAI_scores_ISCO08	public	CC BY 4.0	N
ESCO–O*NET crosswalk and ESCO occupations	esco.ec.europa.eu	public	ESCO terms (European Commission)	N
TalentCorp (2024) GBS impact study — facts with page numbers	talentcorp.com.my	public report	All rights reserved; facts cited, no text reproduced	N
MyCOL 2024/2025 critical occupations — names with pages	mymahir.my / TalentCorp	public brochure	Facts cited, no text reproduced	N
UN SDG target wording	sdgs.un.org	public	UN terms of use	N
Learning resources: 33 courses, 48 videos, 41 articles (links only)	Microsoft Learn, OpenLearn, YouTube, IBM, IFRS, ISO, COSO and others	public	Each provider's terms; linked, not copied	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Synthetic sample CVs (Nur, Hafiz, Mei) and test profiles	created by the team (data/samples, data/eval)	synthetic	Our own (Apache-2.0)	N (fictional people, labelled)
40 labelled job-title queries for the RunPod evaluation	created by the team (data/eval/occupation_titles.yaml)	synthetic	Our own (Apache-2.0)	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Claude Code (Anthropic; Claude Opus models)	Code, tests, documentation, design changes, research summaries, Bahasa Malaysia interface strings (native review pending)	major
OpenAI Codex	Code, tests, UI redesign, pipeline verification	major

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Every change went through small commits that we reviewed and can explain. 175 automated tests, ruff, mypy and lint ran before each commit, and screens were checked in a browser against the real API. Inside the product, the language model never produces numbers: code computes them, quotes are checked against the CV, and explanations cite computed facts. Every course, video and article was verified (link, title, channel or publisher) before inclusion. We tested the app ourselves end to end and redesigned what failed.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☒ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	credits: Autoscale + Scheduled deployments, PostgreSQL	Hosting the live prototype, its production database and the daily data pipeline
RunPod	credits: Serverless GPU endpoint	Serverless GPU endpoint serving BGE-M3 and the BGE reranker

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit is essential to the deployed prototype as built: it runs the web app, its database and the daily data collection, but any Python host with PostgreSQL could replace it. RunPod powers the matching quality: on our 40-title test, top-1 accuracy is 72% with the GPU models and 58% without, and Malay titles drop from 75% to 0%. The app keeps working without RunPod, because every call has a time budget and a labelled keyword fallback, but multilingual matching depends on it.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

Only synthetic data is used; sample CVs are fictional and labelled. There are no accounts or logins. When a CV is uploaded, it is read in memory; names, IC numbers, phone numbers and emails are removed before anything else happens, and only the scrubbed text is sent to OpenAI. Job titles and task descriptions are sent to our RunPod endpoint. The stored profile keeps skill claims, their quoted CV sentences, weekly hours and preferences, under a random ID kept in the user's browser; the full CV text and any name are not stored. A name typed for the PDF CV stays in the browser. Data sits in Replit's PostgreSQL; API keys are environment secrets, never in code. Admin actions need a token. Records are kept for the hackathon period; there is no deletion feature yet. Protected attributes never change a result (paired-profile test).

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Misreading or inventing: gpt-4o-mini can misread a CV. Every claim must quote the CV and code rejects quotes that are not there; levels stay editable self-claims. Explanations may not contain numbers and must cite computed facts.
Bias: protected attributes are removed before matching, and a test checks that CVs differing only in name, gender, religion, race and age give identical results.
Provider dependency and cold starts: RunPod can take about 20 seconds to start. Calls use a queue and a time budget, with a labelled BM25 fallback; gpt-4o-mini has a template fallback.
Evidence limits: AI-use data reflects global Claude conversations, not Malaysian workplace adoption; exposure is technical potential, not job loss. Each is labelled.
Scope: full role plans exist for GBS finance only; scenario weights and projected fit are declared assumptions and estimates.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☒ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☒ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☒ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☒ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☒ Any use of official SDC sponsor resources has been declared in Section 5.
- ☒ The prototype described here is accessible to judges through the links provided in Section 1.
- ☒ All fields in this template have been completed within the word limits stated.

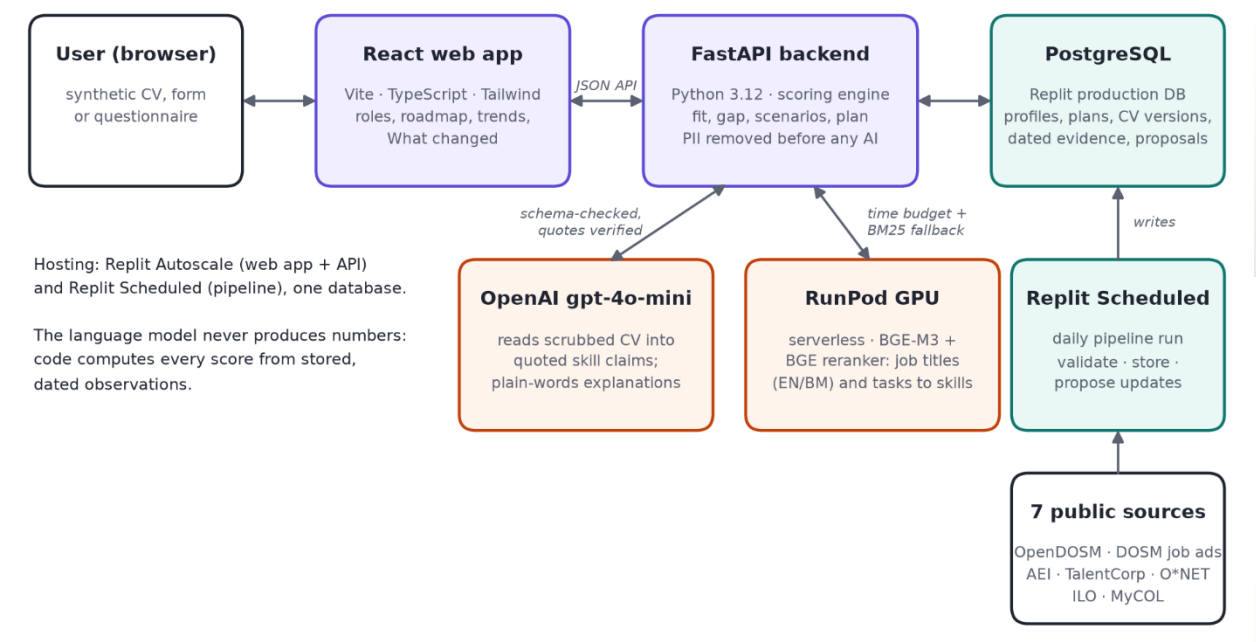
8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Kydyrgaliyev Meiirman, Team Leader

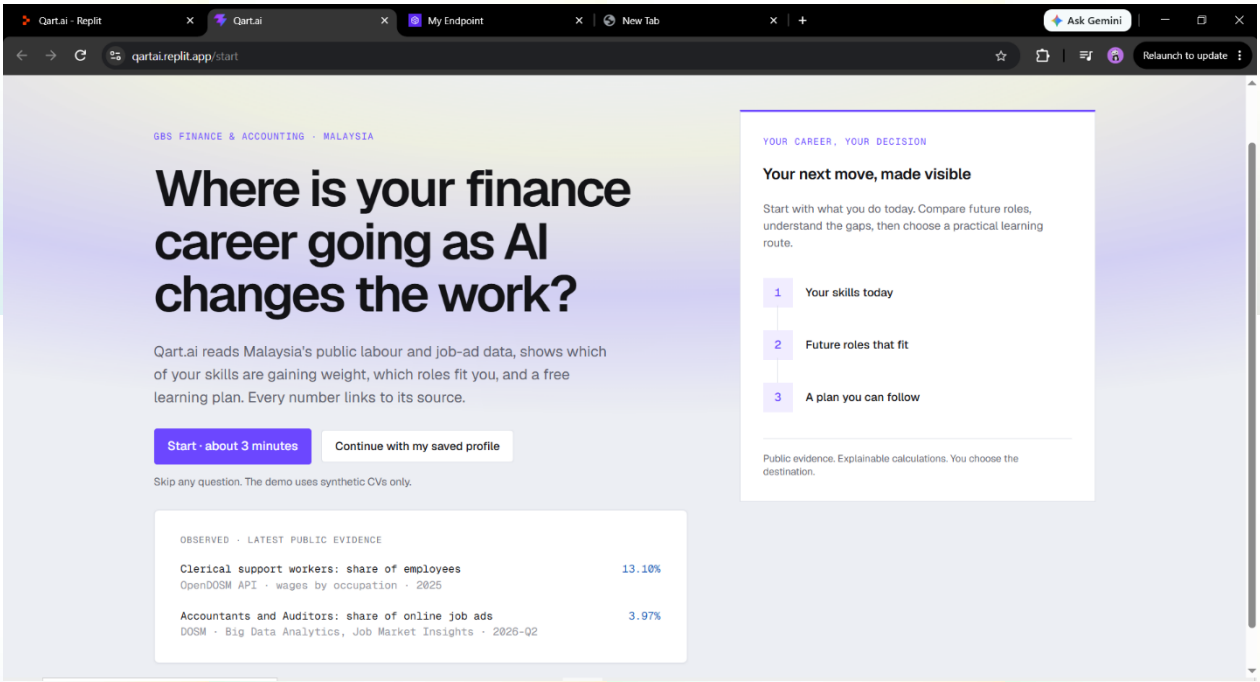
25 October 2026

Appendix A: System block diagram

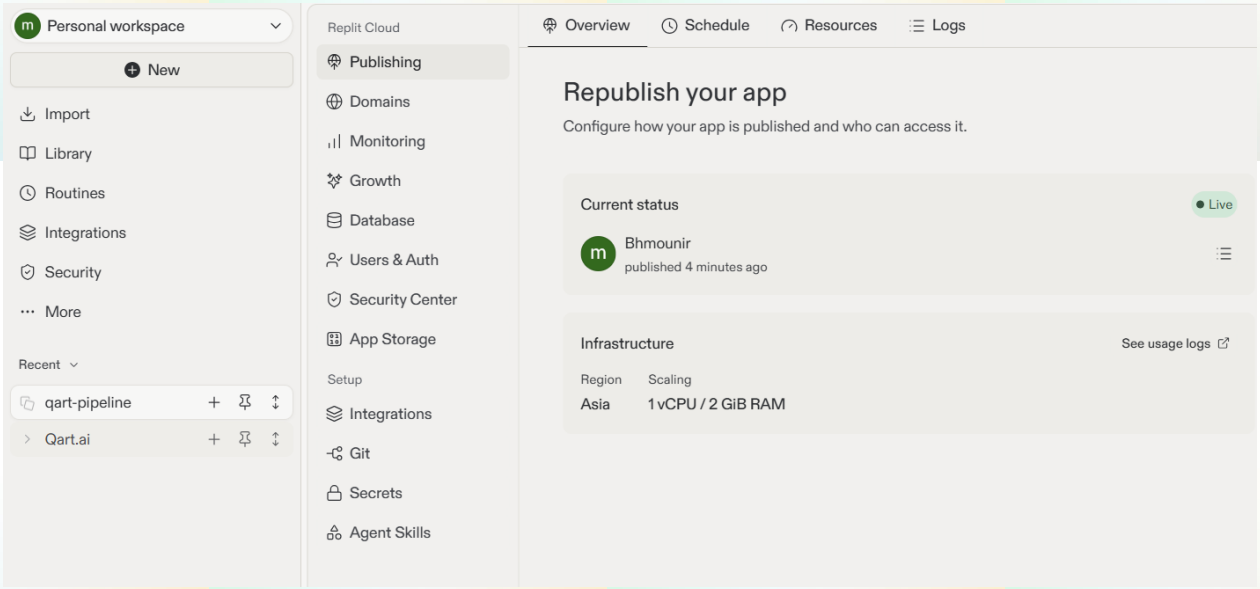


How the components connect end to end (see Section 2.1).

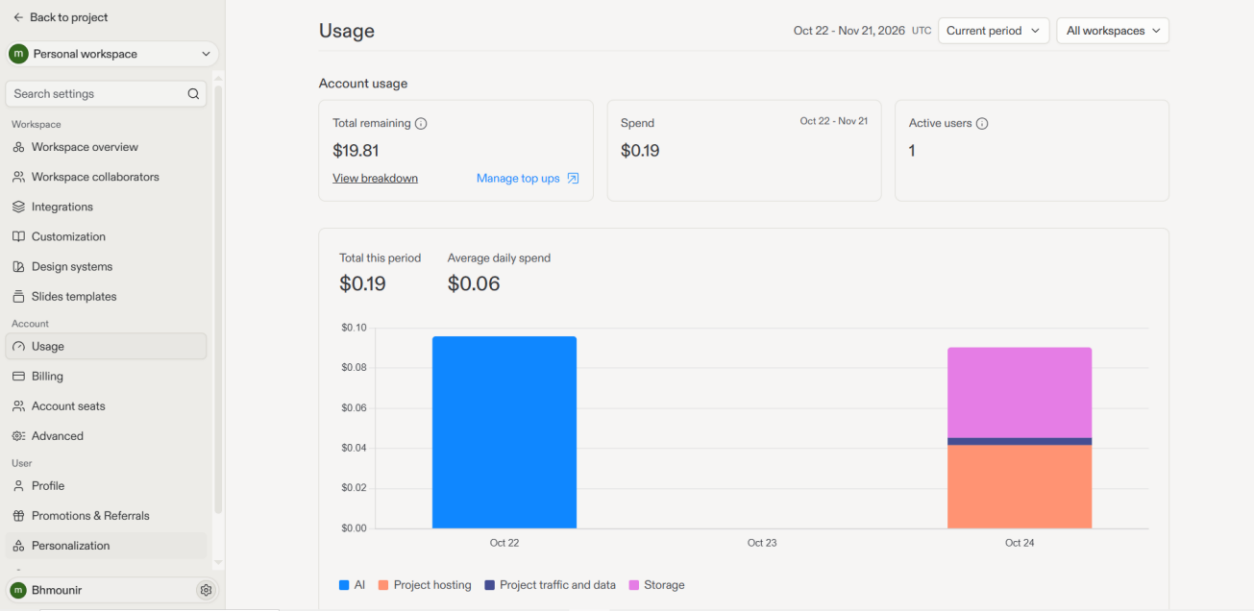
Appendix B: Proof of Replit and RunPod use



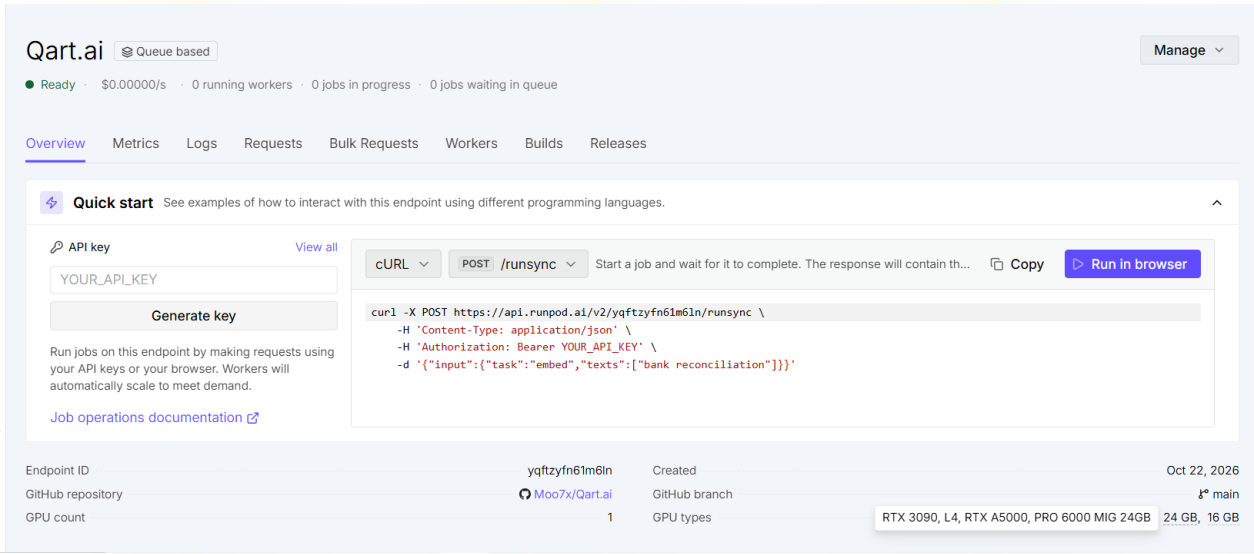
Replit: the live prototype at qartai.replit.app/start



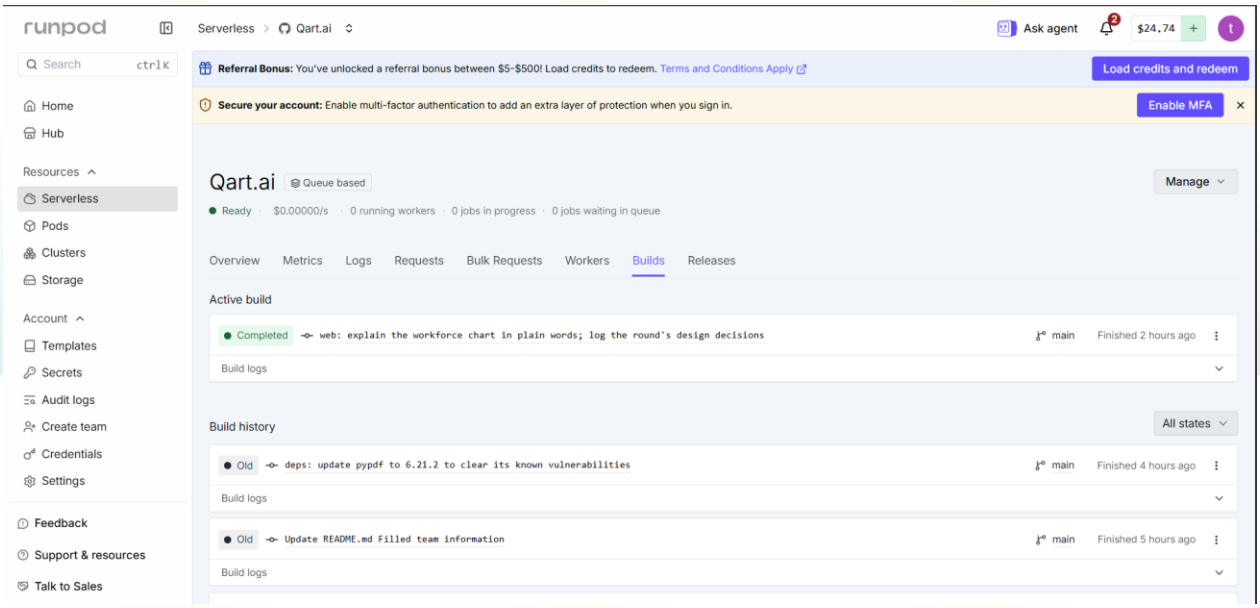
Replit: the Scheduled deployment (qart-pipeline) live, Asia, 1 vCPU / 2 GiB



Replit: account usage, credits being used (AI, hosting, traffic, storage)



RunPod: the Qart.ai serverless endpoint, Ready, built from GitHub main, GPU pools



RunPod: builds from the GitHub repository, and the account credit balance